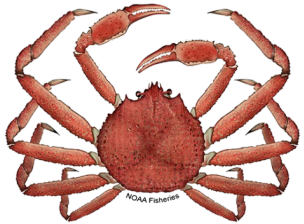
The How and the Why:

Molecular insights into ocean acidification responses in Alaskan crabs



Shellfish Seminar Series, AFSC Kodiak Lab (remote), March 2026

Laura Spencer, NOAA Affiliate at Univ. of Washington
Chris Long, NOAA AFSC Kodiak
Jennifer Gardner, NOAA AFSC Kodiak
Ingrid Spies, NOAA AFSC Seattle
Steven Roberts, University of Washington SAFS

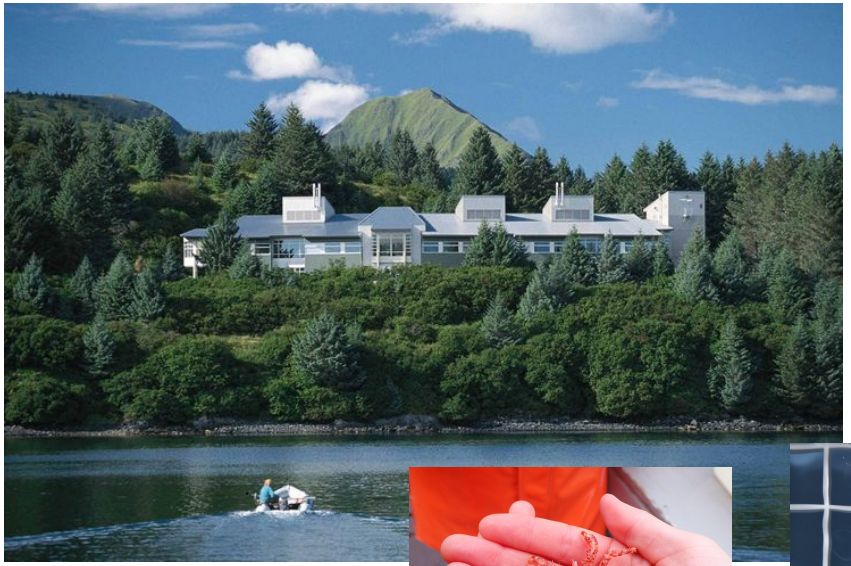
SCHOOL OF AQUATIC & FISHERY SCIENCES

CICOES

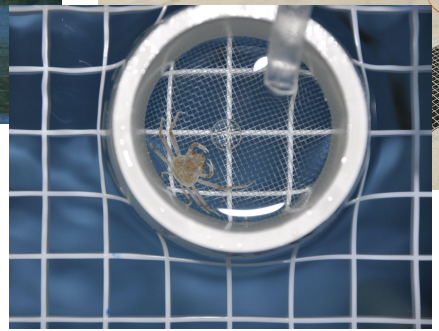
NOAA FISHERIES

OAP

Integrating molecular tools with experimental stress-response studies









NOAA Kodiak
Research Station

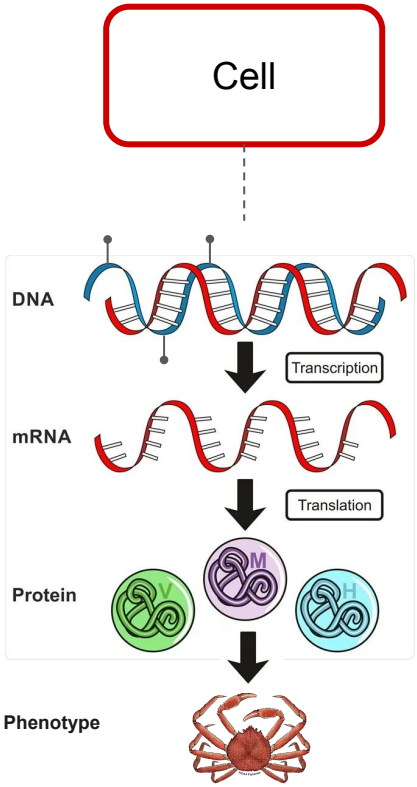


Today:

1. What is molecular data, how do we get it, & what can it tell us?

2. Examples of how we're using in two Ocean Acidification studies
    a. Snow crab juveniles
    b. Tanner crab adults (males)

What is molecular data?



Adapted from: udaix/Shutterstock.com

Appr. genome sizes (base pairs)

Snow: 2.0 billion
Tanner: 1.7 billion
Red king crab: 3.8 billion
Dungeness: 660 million
Pacific cod: 556 million
Humans: 3.1 billion

E.g. DNA sequence data - Who are they?

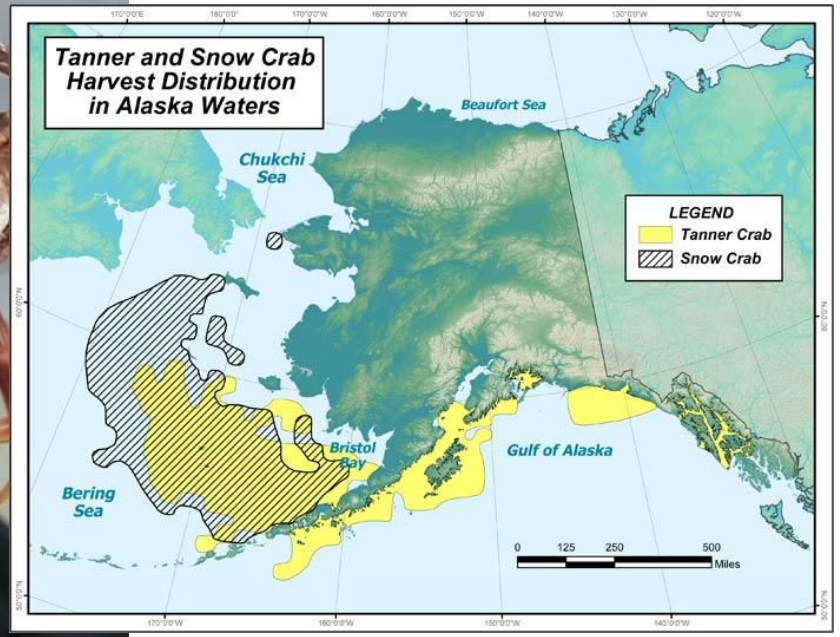
TGCCGGGTCCAGTTCGCTGGGCTTTCGGCGCCTCGAGCTGACGGAAAGAT
TGCTGGGTCCAGTTCGCTGGGCTTTCGGCGCCTCGAGCTGACGGAAAGAT

E.g. RNA sequence data - What are they doing right now?

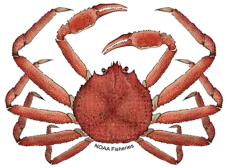
UGCCGGGUCCAGUUCGCUGGGCUUUCGGCGCCUCGAGCUGACGGAAAGAU

Snow & Tanner crabs, *Chionoecetes spp.*

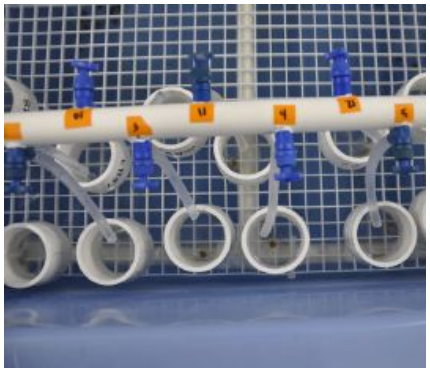




ADFG



Experiment: What are the impacts of OA on snow crab juveniles?



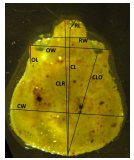


Laboratory Experiment

- Wild juveniles collected during NOAA 2021 EBS survey
- Held for 396 days, 3 pH treatments (at 4°C) in Kodiak lab

Data:

1. Survival, molting, growth, morphometrics
2. Gene expression (RNASeq), 8hr & 88 days

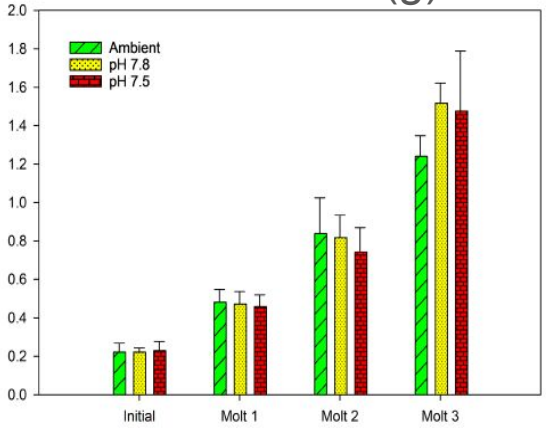


n=30 / treatment

Control	Moderate OA	Severe OA
pH 8.0	*pH 7.8*	*pH 7.5*

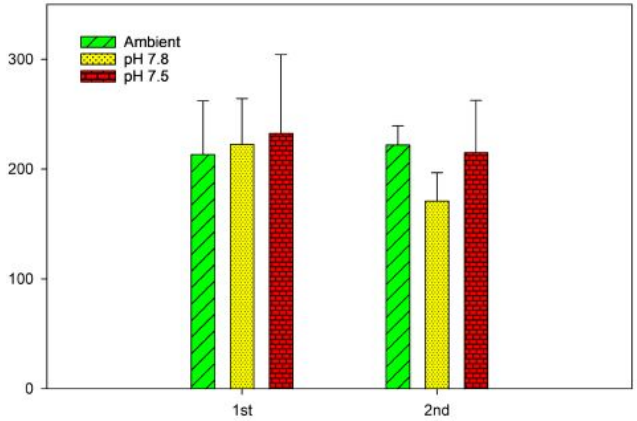
Snow crab juveniles are surprising tolerant of long-term OA

Wet mass (g)



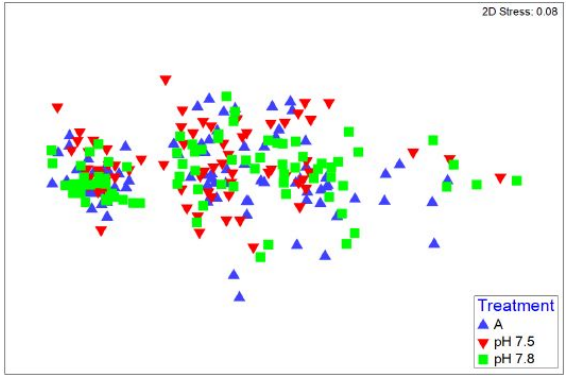
Molt stage

Intermolt duration



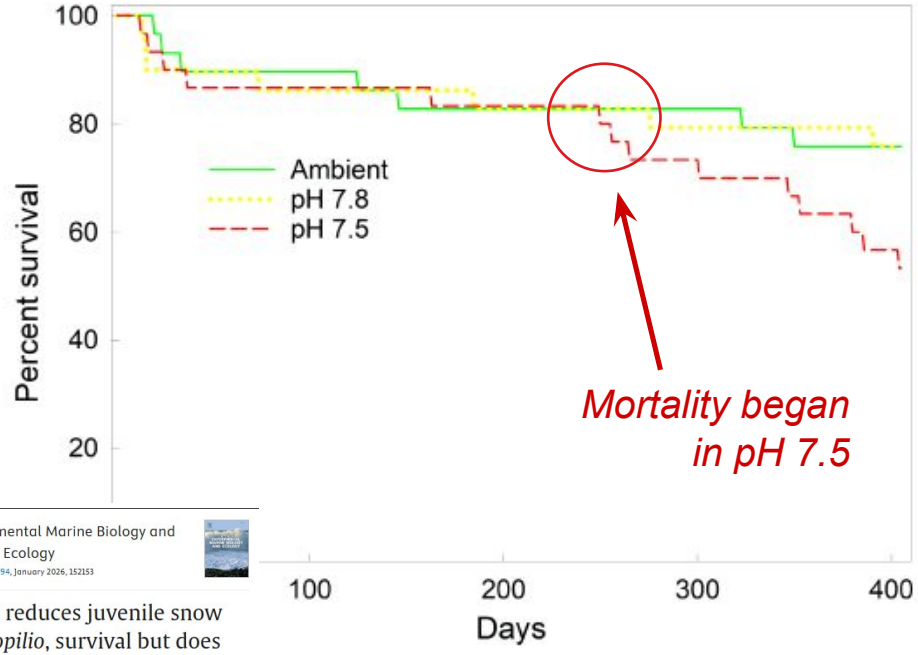
Intermolt

Morphometrics NMDS



No difference in growth, morphometrics over 396-day exposure

Snow crab juveniles are surprising tolerant of long-term OA




Journal of Experimental Marine Biology and Ecology
Volume 594, January 2026, 152153

Ocean acidification reduces juvenile snow crab, *Chionoecetes opilio*, survival but does not affect growth or morphometrics

W. Christopher Long

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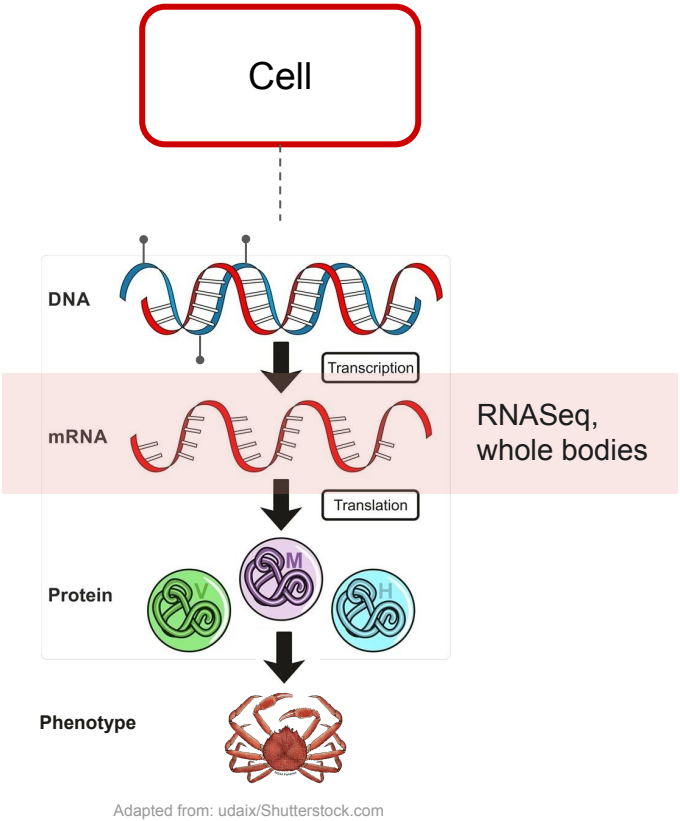
How do snow crab tolerate OA?

Why higher mortality in pH 7.5?



Survival not affected until day 250

Gene expression analysis used to understand OA tolerance

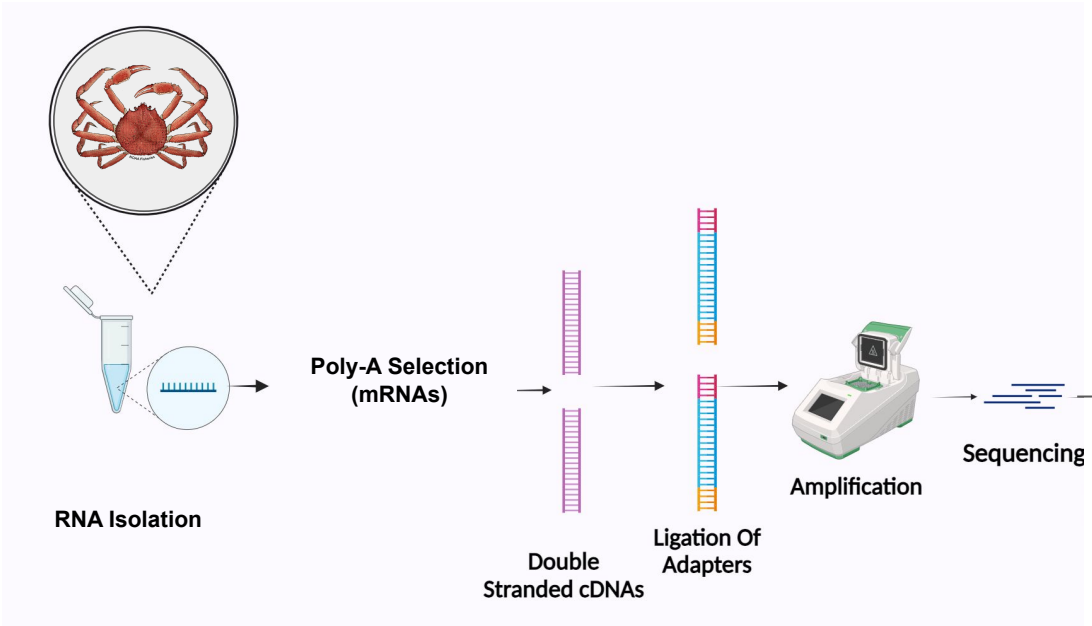


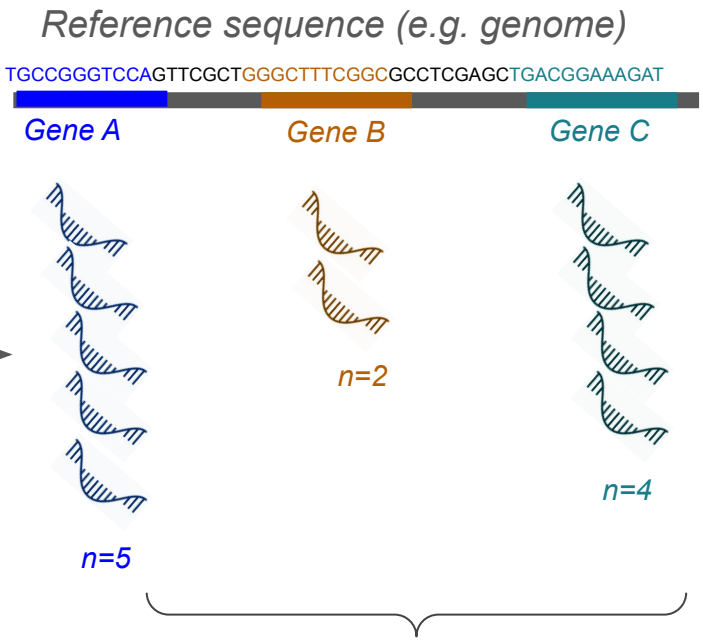
RNA-Seq - approach that measures all mRNA transcripts, ie. expressed genes

Captures shifts in energy allocation

Identify genes, processes important to responding to a stressor

How do we get gene expression data, and what does it look like?





Adapted from an image by Shraddha Tomar, BioRender

Gene count matrix

How do we get gene expression data, and what does it look like?

Samples

Treatment *Control*

Genes

gene_gadmor	PCG001	PCG004	PCG011	PCG015	PCG017	PCG020	PCG029	PCG035
ND1	8276	8202	8327	7750	5508	4351	5673	6385
ND2	39142	43297	57032	31681	33489	30275	38921	36460
COX1	555463	631876	917827	551062	378628	403956	438595	455514
COX2	315625	309958	493176	294979	189346	229132	227614	219894
ATP6	82892	105415	89100	73253	77770	58061	86703	78859
COX3	165275	189321	235193	135559	112870	81377	135462	126615
ND3	10710	13595	10404	7013	10244	2567	9792	8231
ND4L	19364	31196	42855	16261	20566	17877	21021	16269
ND4	37081	42648	67118	30275	29274	37939	35681	34155
ND5	16299	21803	2823[illegible]			19960	19262	18187
ND6	4102	3787	318[illegible]			1774	3772	3391
CYTB	176843	211635	277395			114784	137070	144369
LOC115539476	260	228	333	219	270	266	312	266
LOC115539709	890	849	919	777	561	461	945	1000
LOC115538781	586	596	576	630	417	450	704	730
abhd14a	1197	1381	1629	1031	774	757	1158	1108
acy1	1626	1244	1670	1245	1224	1386	1566	1642
LOC115537228	2106	2402	2555	2008	1845	2826	2127	2235
LOC115537019	659	544	911	564	505	611	696	444
LOC115538651	727	674	630	564	479	599	554	523
LOC115538267	57	192	81	153	104	137	110	104
kbtbd12	1003	743	646	766	715	875	301	216

Gene counts

How do we get gene expression data, and what does it look like?

Samples

Treatment

Control

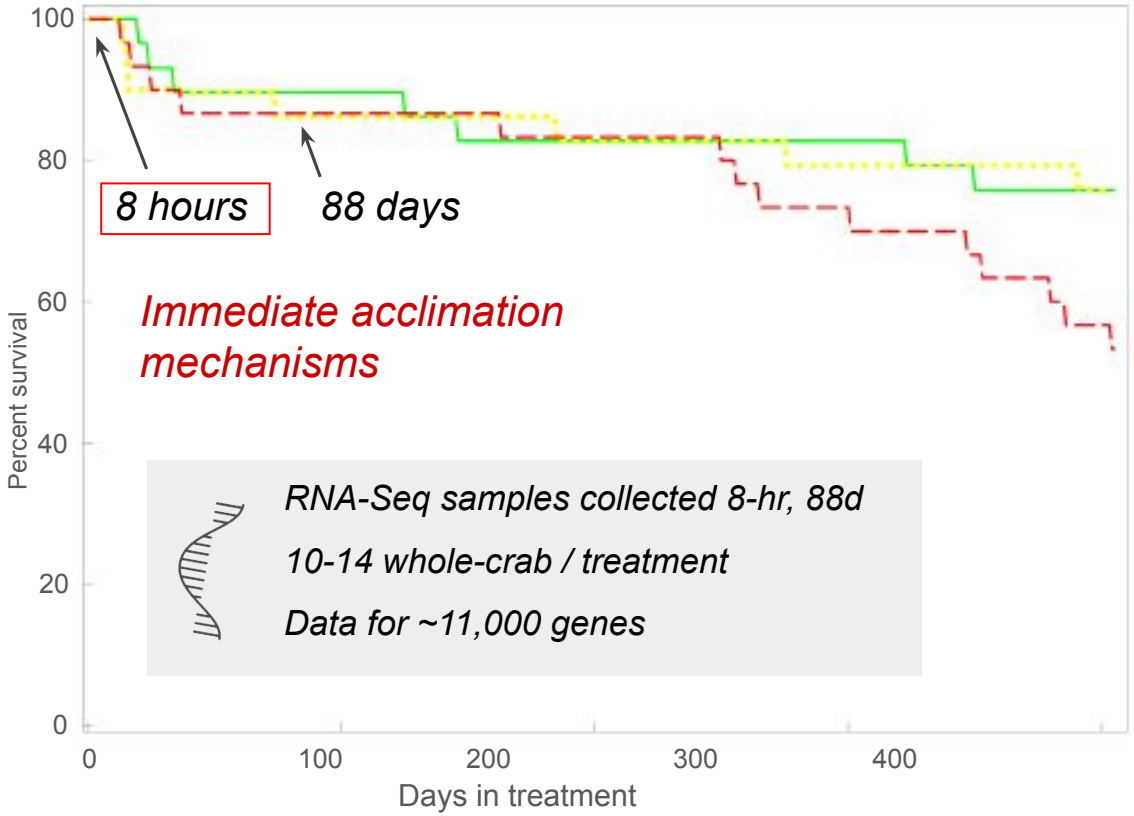
functional info for many gene

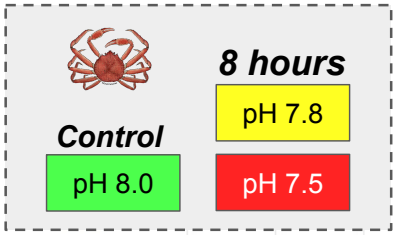
Genes

gene_gadmor	PCG001	PCG004	PCG011	PCG015	PCG017	PCG020	PCG029	PCG035	spid	species	evalue	protein_names
ND1	8276	8202	8327	7750	5508	4351	5673	6385	P55779	GADMO	0.00e+00	NADH-ubiquinone oxidoreductase chain 1 (EC 7.1.1.2) (NA
ND2	39142	43297	57032	31681	33489	30275	38921	36460	P55780	GADMO	2.40e-158	NADH-ubiquinone oxidoreductase chain 2 (EC 7.1.1.2) (NA
COX1	555463	631876	917827	551062	378628	403956	438595	455514	Q36775	GADMO	0.00e+00	Cytochrome c oxidase subunit 1 (EC 7.1.1.9) (Cytochrome c
COX2	315625	309958	493176	294979	189346	229132	227614	219894	Q37741	GADMO	1.12e-132	Cytochrome c oxidase subunit 2 (EC 7.1.1.9) (Cytochrome c
ATP6	82892	105415	89100	73253	77770	58061	86703	78859	P55778	GADMO	2.60e-104	ATP synthase subunit a (F-ATPase protein 6)
COX3	165275											C 7.1.1.9) (Cytochrome c
ND3	10710											e chain 3 (EC 7.1.1.2) (NA
ND4L	19364											e chain 4L (EC 7.1.1.2) (NA
ND4	37081											e chain 4 (EC 7.1.1.2) (NA
ND5	16299											e chain 5 (EC 7.1.1.2) (NA
ND6	4102											e chain 6 (EC 7.1.1.2) (NA
CYTB	176843											t 3) (Complex III subunit
LOC115539476	260											g protein 1 (Protein 123F
LOC115539709	890											sion 1 protein) (Fus-1 pr
LOC115538781	586											35) (Hyaluronoglucosam
abhd14a	1197	1381	1629	1031	774	757	1158	1108	Q1LV46	DANRE	4.34e-34	Protein ABHD14A (EC 3.-.-.-) (Alpha/beta hydrolase domai
acy1	1626	1244	1670	1245	1224	1386	1566	1642	Q6PTT0	RAT	2.57e-16	Aminoacylase-1B (ACY-1B) (EC 3.5.1.14) (ACY IB) (N-acyl-L-
LOC115537228	2106	2402	2555	2008	1845	2826	2127	2235	Q6PHS9	MOUSE	1.07e-34	Voltage-dependent calcium channel subunit alpha-2/delta
LOC115537019	659	544	911	564	505	611	696	444	Q90339	CYPCA	1.94e-172	Myosin heavy chain, fast skeletal muscle
LOC115538651	727	674	630	564	479	599	554	523	Q9NXG6	HUMAN	5.61e-20	Transmembrane prolyl 4-hydroxylase (P4H-TM) (EC 1.14.11
LOC115538267	57	192	81	153	104	137	110	104	Q8CIW6	MOUSE	3.91e-23	Solute carrier family 26 member 6 (Anion exchange transp
kbtbd12	1003	743	646	766	715	875	301	216	Q3ZB90	DANRE	0.00e+00	Kelch repeat and BTB domain-containing protein 12 (Kelch

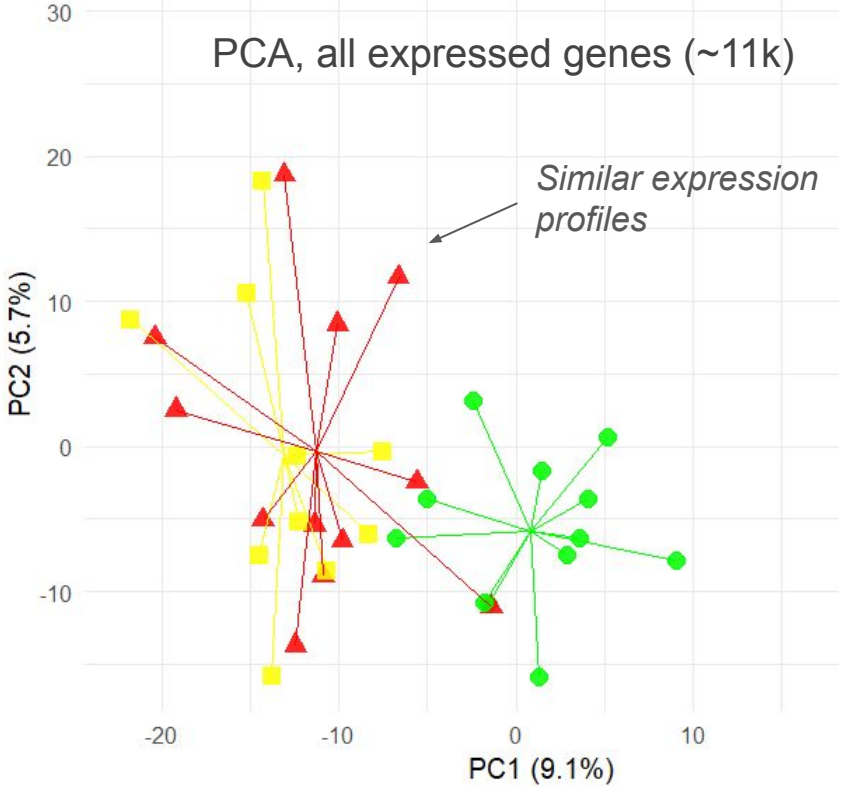
Common analysis: ***compare gene counts among treatments***
using GLMs, negative binomial distribution

Gene expression analysis used to understand OA tolerance





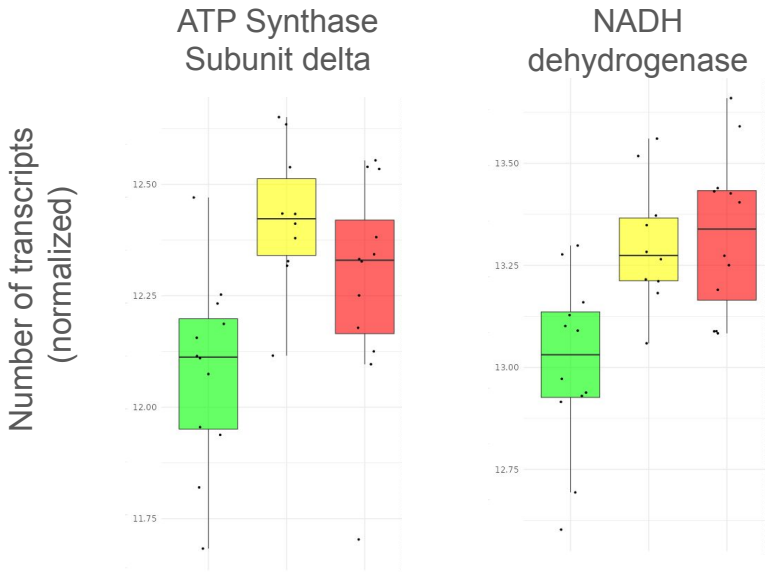
Immediate OA acclimation mechanisms



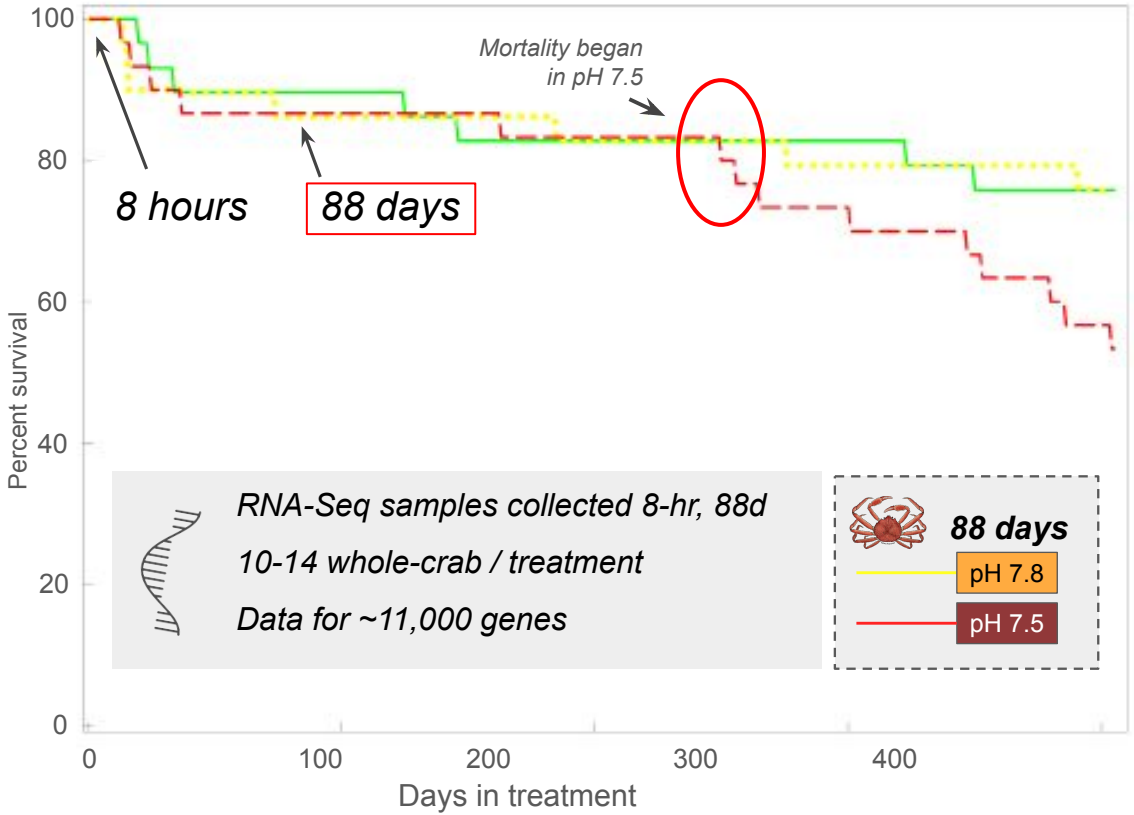
Mitochondrial genes highly active!

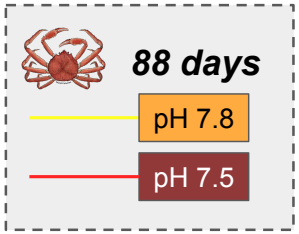
- Energy production
- Damage mitigation



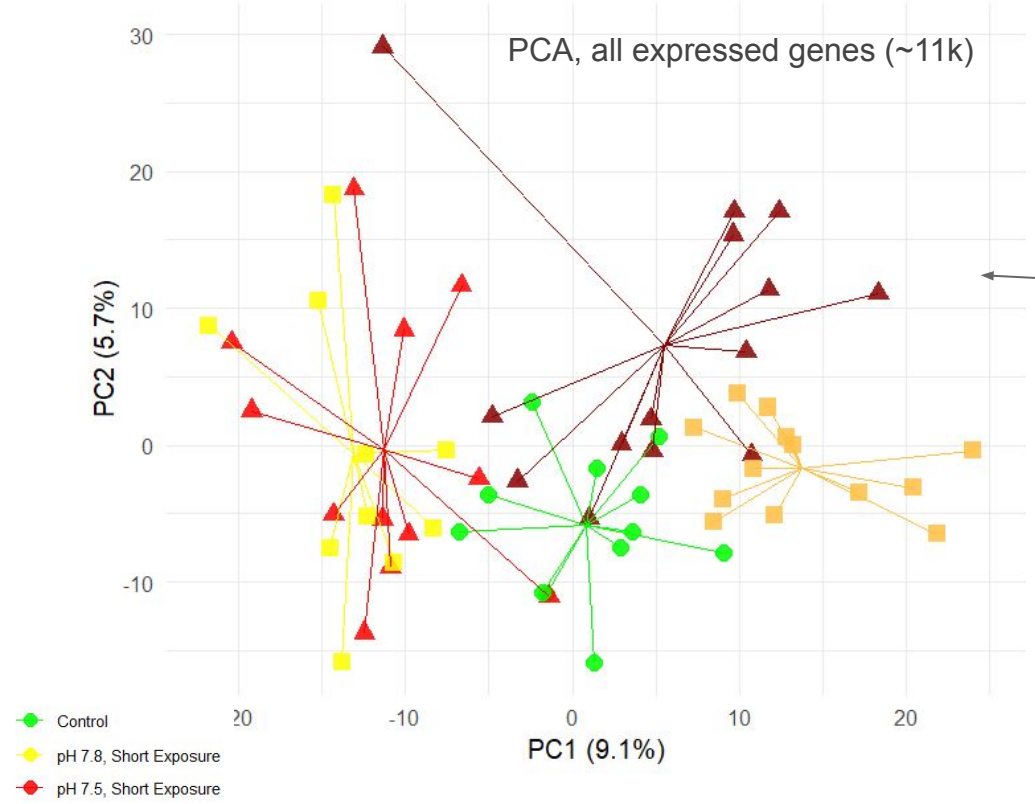


Gene expression analysis used to understand OA stress & tolerance



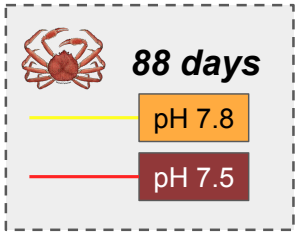


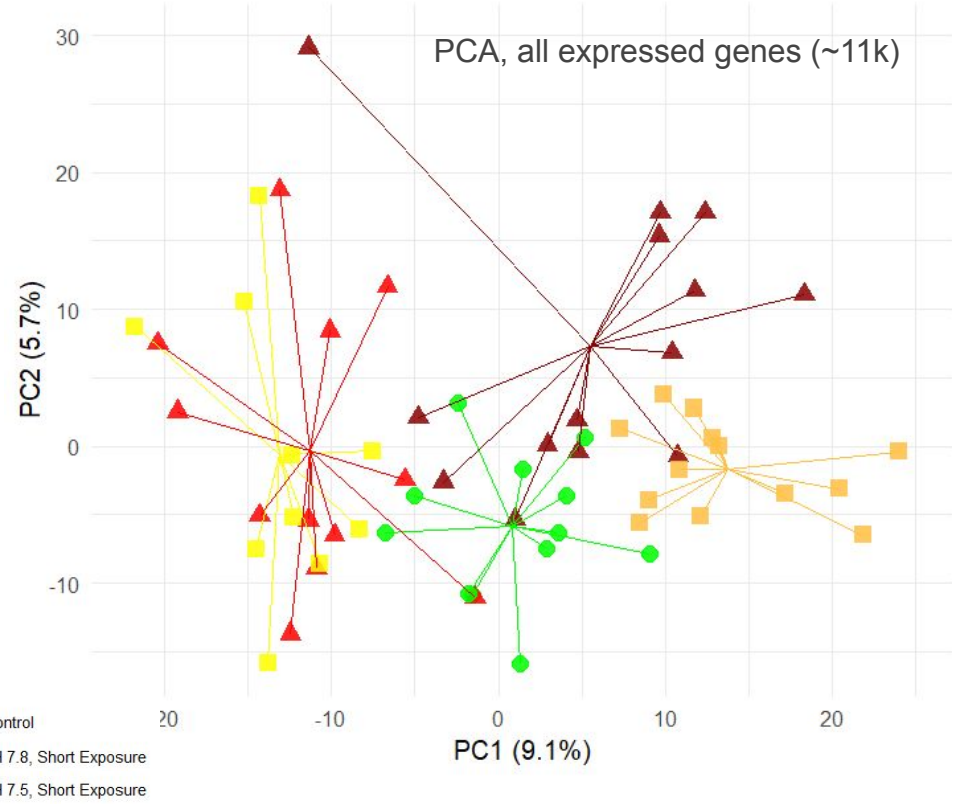
Chronic OA stress indicators

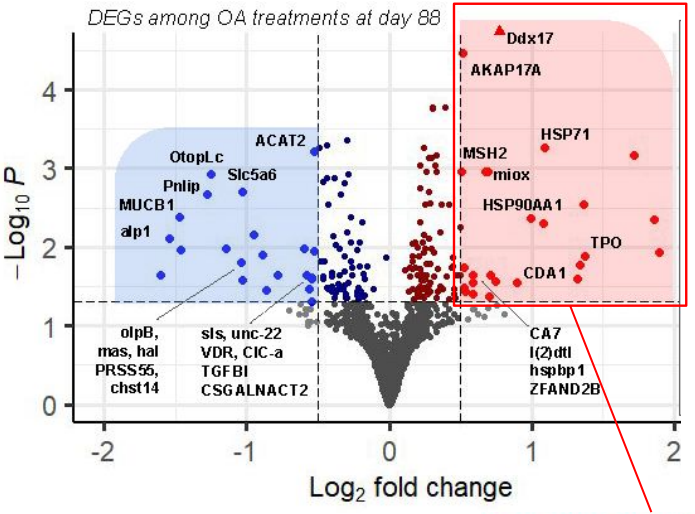


Compare expression of OA treatments at 88 days

Chronic OA stress indicators





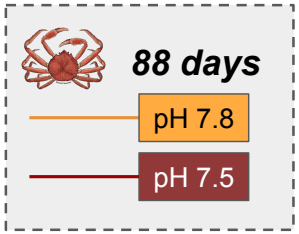


Stress response, protein folding
Heat shock protein 1 (HSP71)
Heat shock protein HSP 90-alpha (HSP90AA1)
Hsp70-binding protein 1 (Hspbp1)
AN1-type zinc finger protein 2B (ZFAND2B)

Chitin catabolism
Chitin deacetylase 1 (CDA1)

Acid-base balance, redox
Thyroid peroxidase (TPO)
Carbonic anhydrase 7 (CA7)

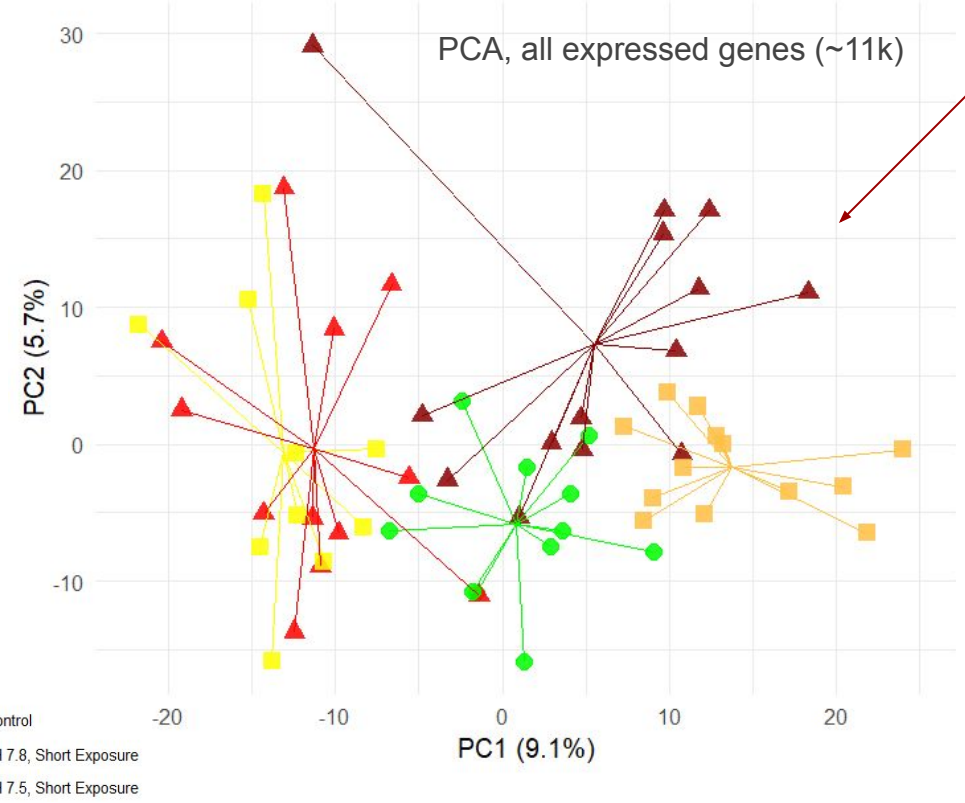
Genome maintenance
ATP-dependent RNA helicase (Ddx17, -log(p)=12)
A-kinase anchor protein 17A (AKAP17A)
DNA mismatch repair protein (MSH2)
Protein lethal(2)denticleless (l(2)dtl)

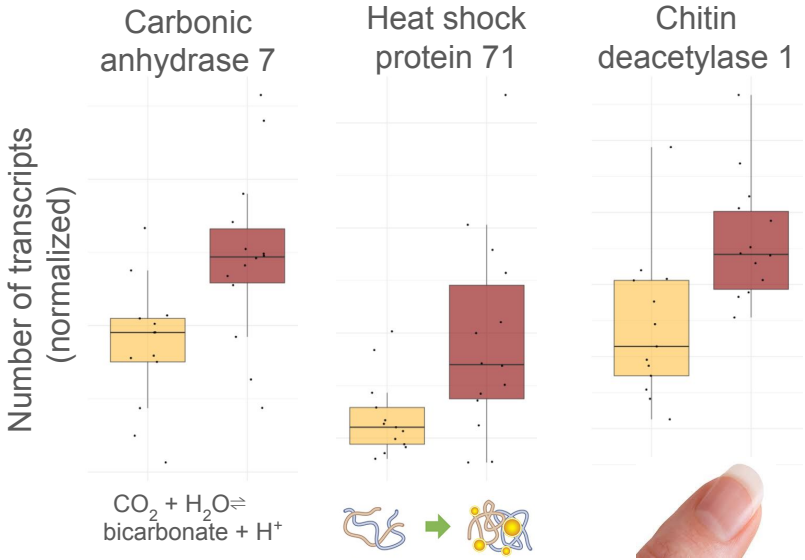


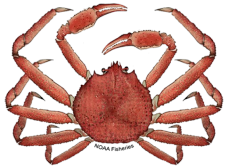
Chronic OA stress indicators

Higher activity, OA-stressed (pH 7.5)

- pH regulation
- Protein & DNA damage repair
- Chitin formation, modification







Results: Short-term mechanisms, long term consequences

How do snow crab tolerate OA?

- Able to sustain energy production & repair mitochondria

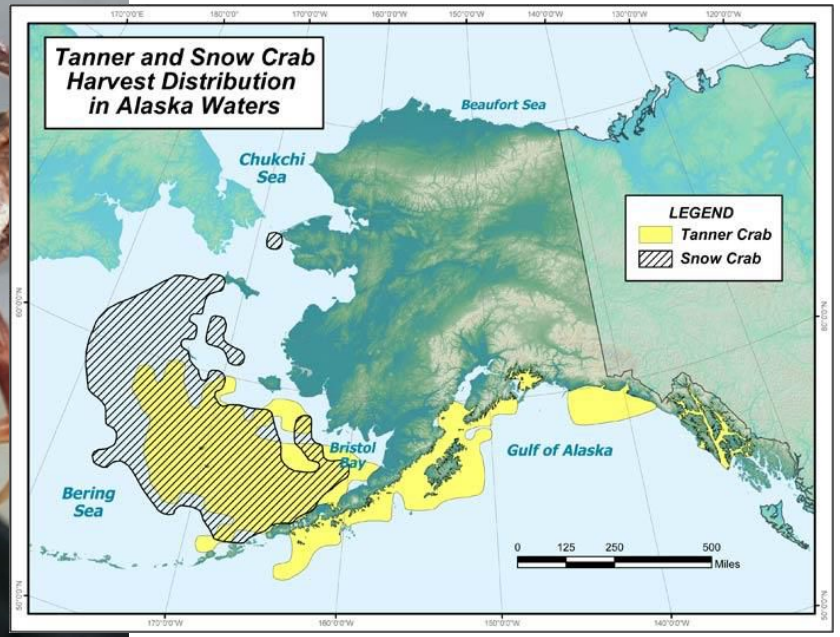
Why higher mortality in pH 7.5?

- Chronic cellular stress mitigation (protein, DNA damage) & chitin maintenance
- Mortality during molting - over-extended stress & damage-response systems

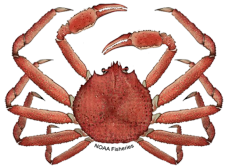
Carbonic anhydrase - could be a useful biomarker of OA stress

Snow & Tanner crabs, *Chionoecetes spp.*

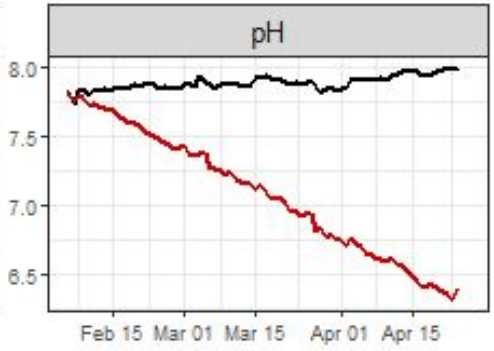


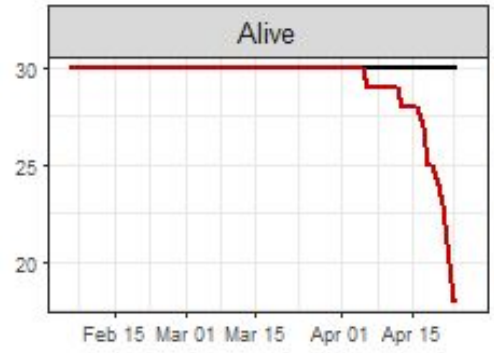


ADFG



Experiment: What is distinct about Tanners that survive severe OA?



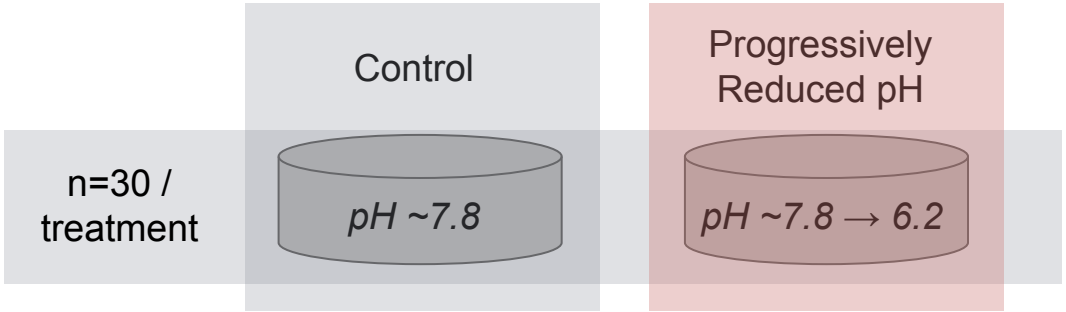


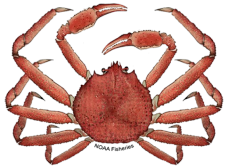
Laboratory Experiment

- Wild adult males collected during NOAA EBS survey
- pH reduced over ~2 month until ~50% mortality

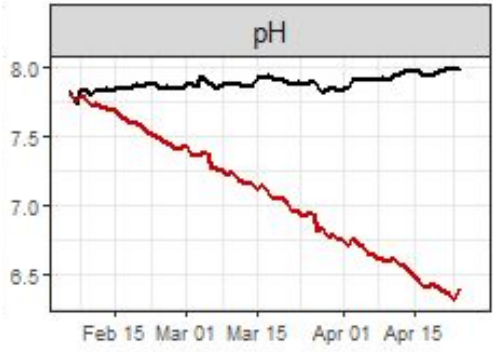
Data:

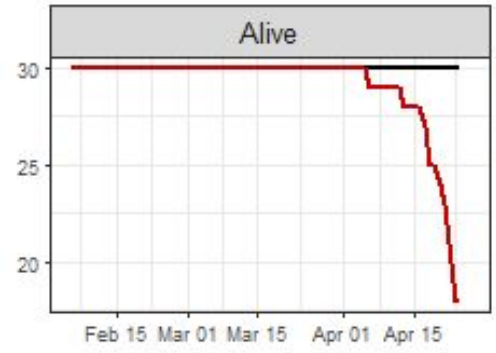
1. Survival, size
2. Genetics (lcWGS) & Gene expression (RNASeq)





Experiment: What is distinct about Tanners that survive severe OA?





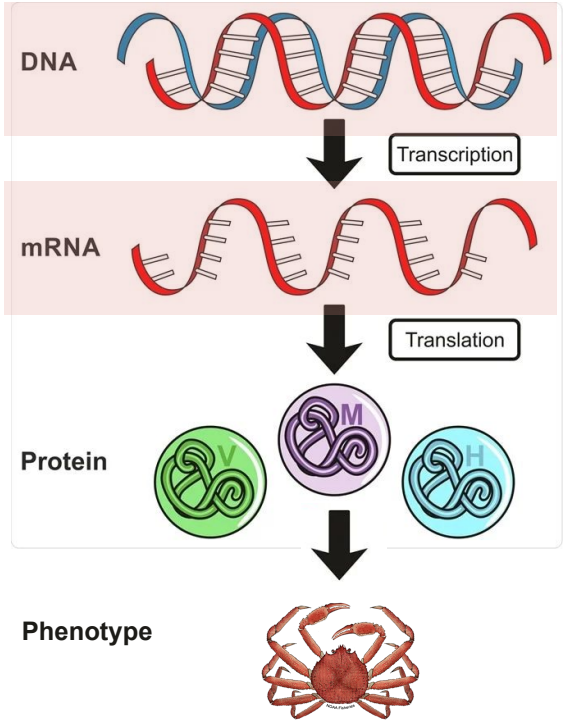
Genetic data (lcWGS)

- Whole-genomes sequenced ~2x

Expression data (RNAseq)

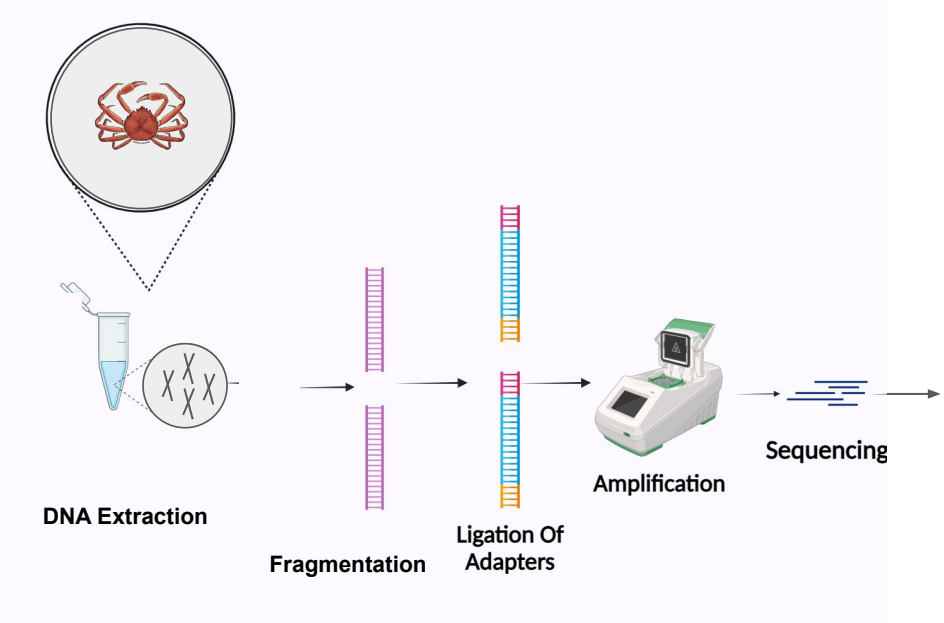
- Gill
- Hepatopancreas
- Muscle

n=14 OA Survivors
n=17 Control



Adapted from: udaix/Shutterstock.com

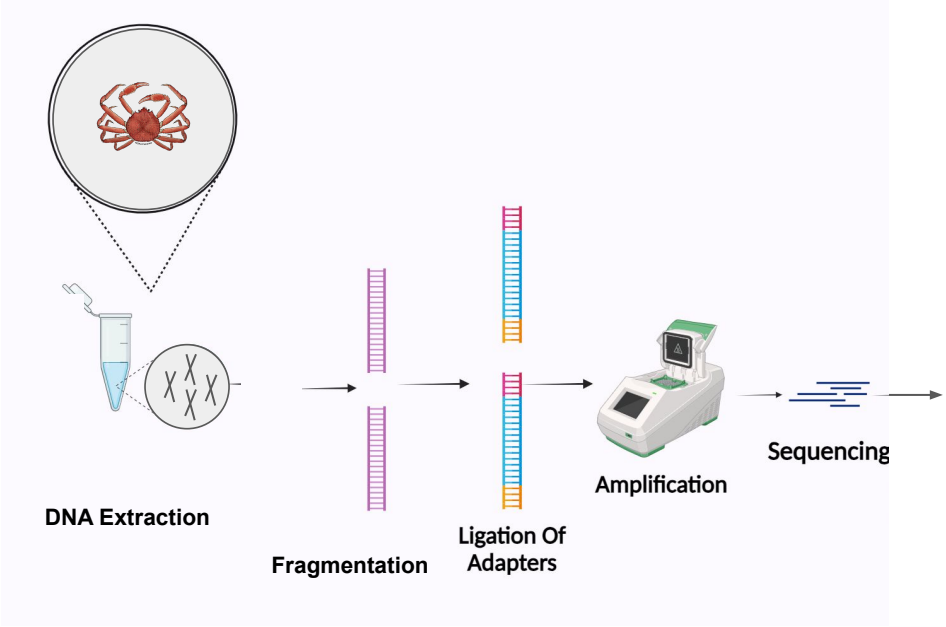
How do we get genetic data, and what does it look like?





Adapted from an image by Shraddha Tomar, BioRender

How do we get genetic data, and what does it look like?



Adapted from an image by Shraddha Tomar, BioRender

Reference sequence (e.g. genome)

TGCCGGGTCCAGTTCGCTGGGCTTTCGGCGCCTCGAGCTGACGGAAAGAT

Individual A: SNP genotype = A/A

TGCCGGGTCCAGTACGCTGGGCTTTCGGCGCCTCGAGCTGACGGAAAGAT
TGCCGGGTCCAGTACGCTGGGCTTTCGGCGCCTCGAGCTGACGGAAAGAT

Individual A: SNP genotype = T/A

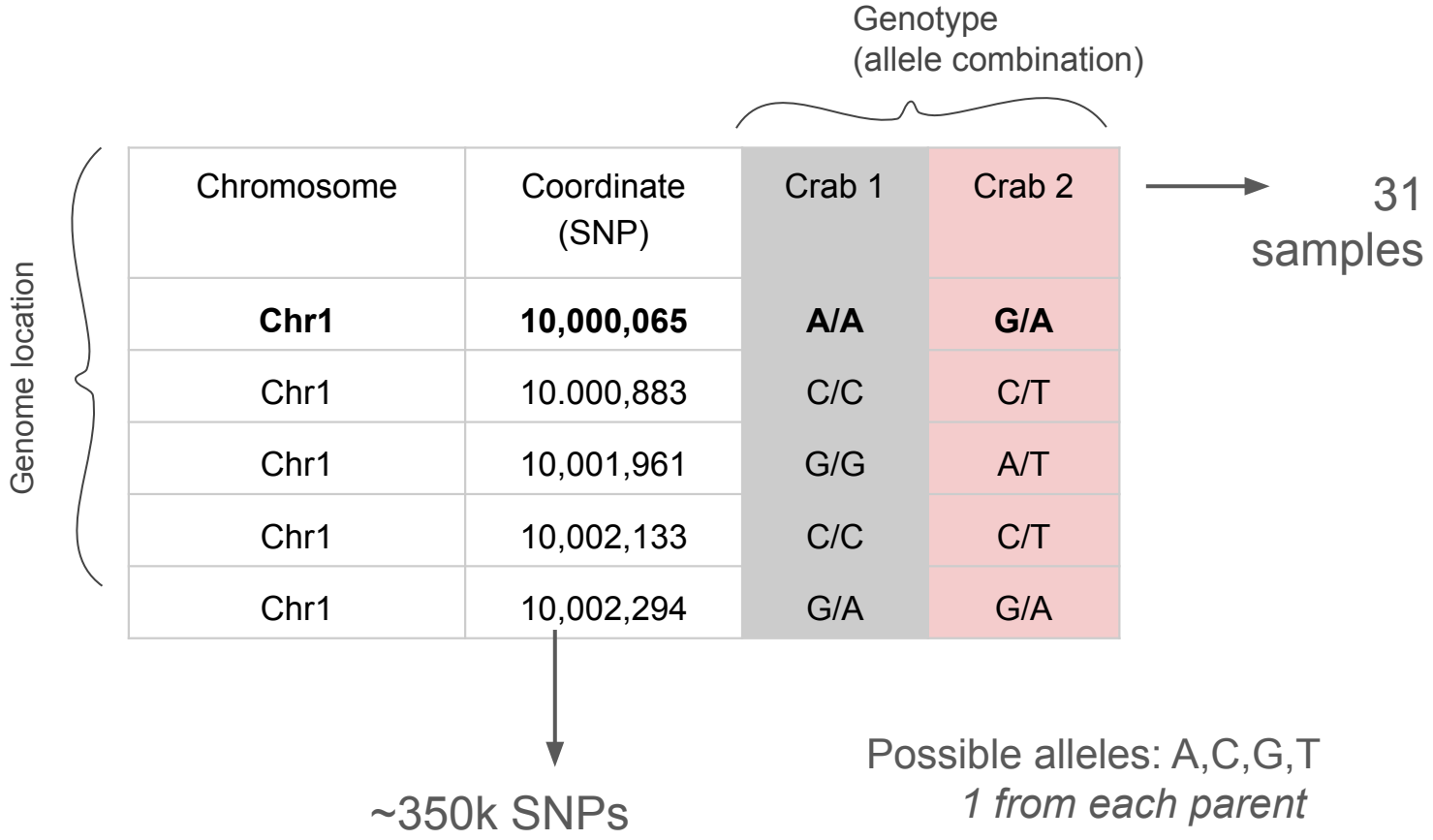
TGCCGGGTCCAGTTCGCTGGGCTTTCGGCGCCTCGAGCTGACGGAAAGAT
TGCCGGGTCCAGTACGCTGGGCTTTCGGCGCCTCGAGCTGACGGAAAGAT

Individual A: SNP genotype = T/T

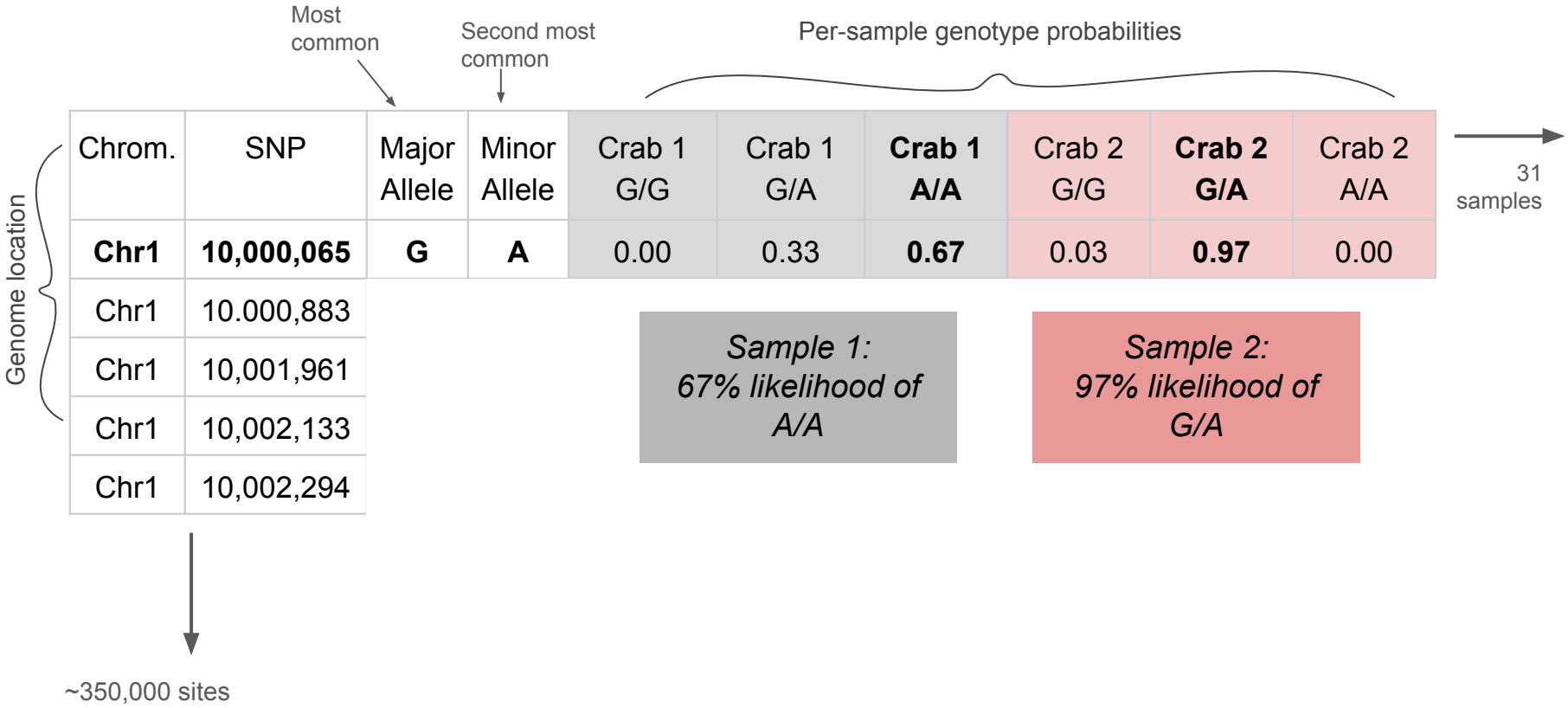
TGCCGGGTCCAGTTCGCTGGGCTTTCGGCGCCTCGAGCTGACGGAAAGAT
TGCCGGGTCCAGTTCGCTGGGCTTTCGGCGCCTCGAGCTGACGGAAAGAT

Single nucleotide polymorphisms (SNPs)

What does genetic data look like?



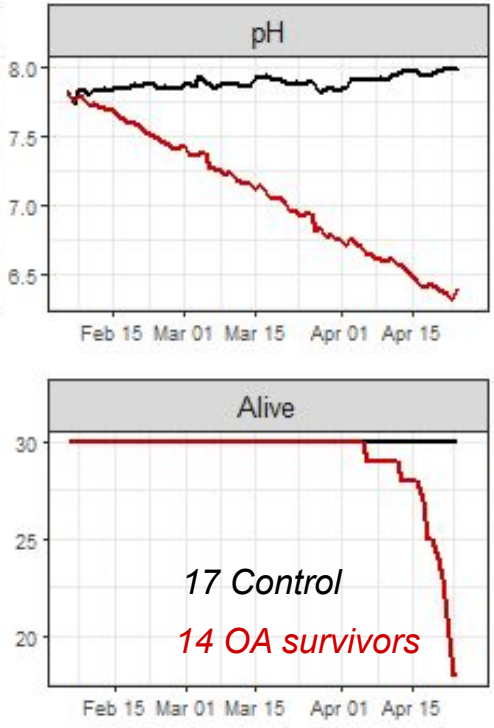
What does genotype likelihood data look like?

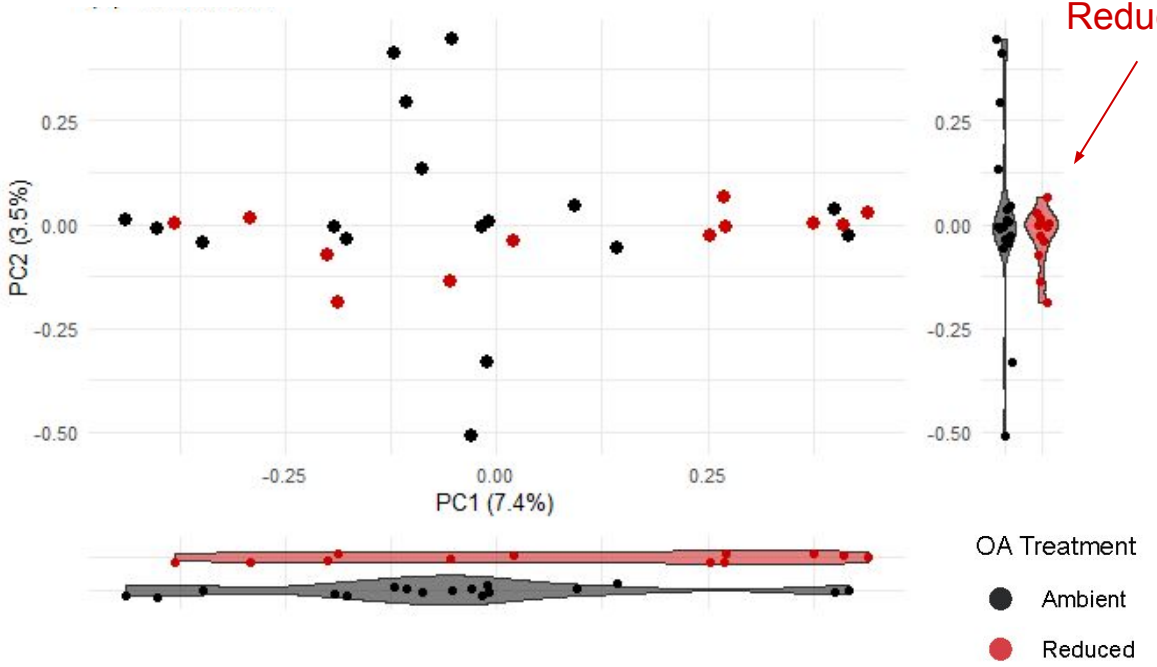


Chrom.	SNP	Major Allele	Minor Allele	Crab 1 G/G	Crab 1 G/A	**Crab 1 A/A**	Crab 2 G/G	**Crab 2 G/A**	Crab 2 A/A
Chr1	**10,000,065**	**G**	**A**	0.00	0.33	**0.67**	0.03	**0.97**	0.00
Chr1	10.000,883								
Chr1	10,001,961								
Chr1	10,002,133								
Chr1	10,002,294								

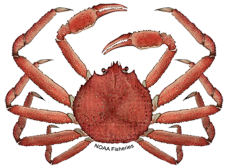
PCA, genetic data (~350k SNPs)

Tanner crab selection study

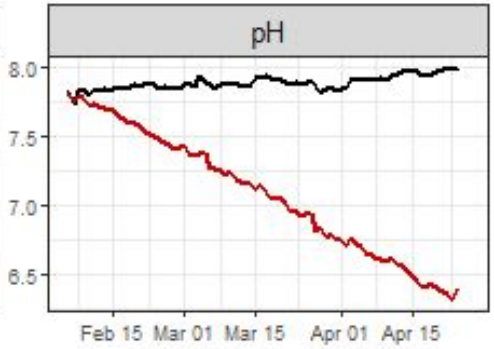


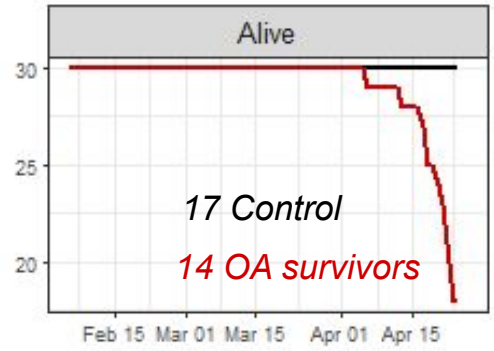


Less variability in Reduced pH



Is there genetic differentiation among OA Survivors & Control?

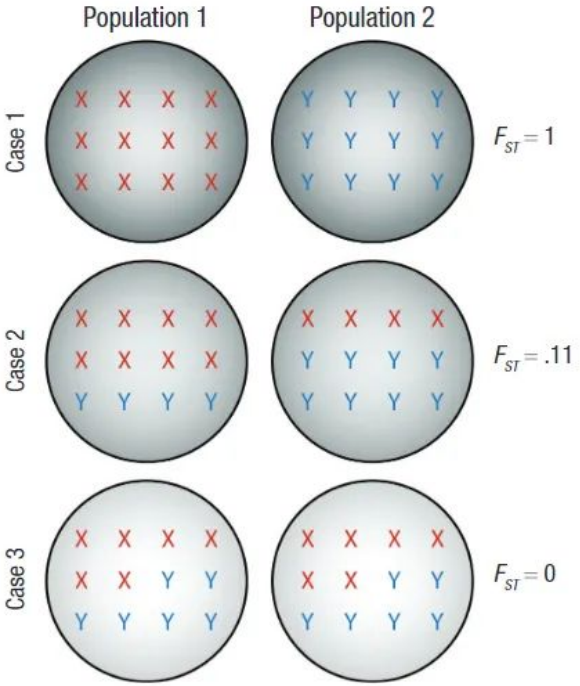


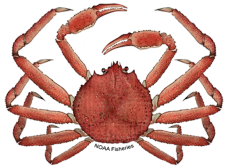


Fixation index (F_{ST}) =

$$\frac{\text{Var(between) - Var(within)}}{\text{Var(between)}}$$

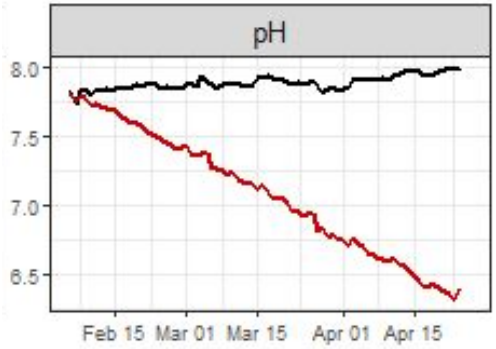
Fst ranges from 0 → 1

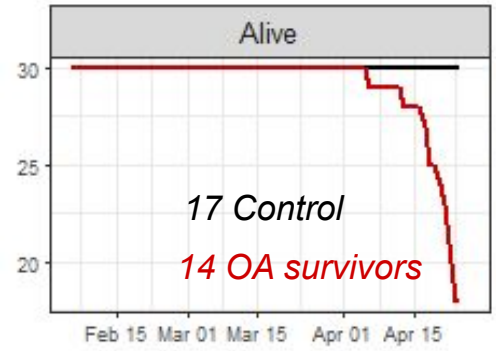


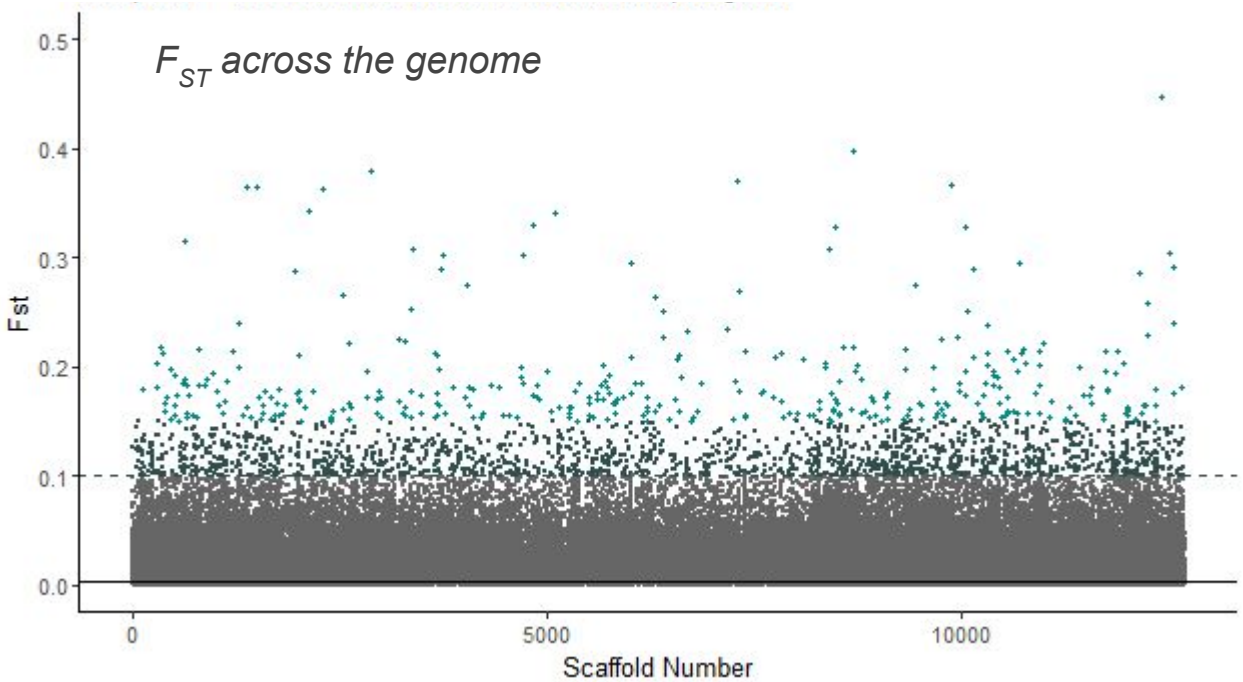


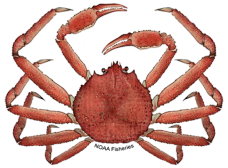
No major genome-wide genetic differences

Mean F_{ST} very low: 0.0027 ± 0.011 SD
Small set of fairly differentiated SNPs (n=~1,900)

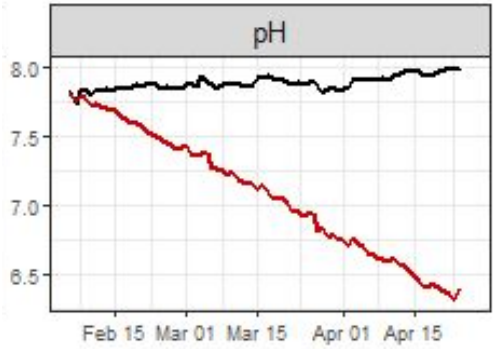


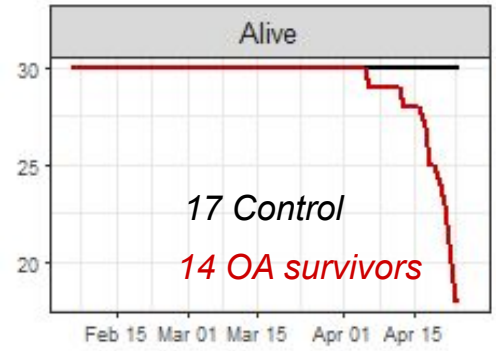




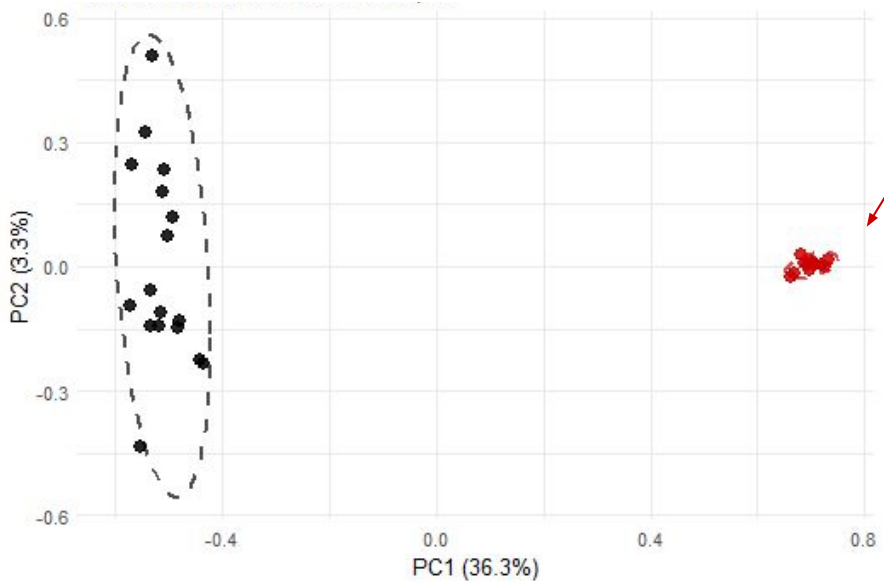


Small set of differentiated SNPs, which had reduced genetic variation in OA survivors





PCA conducted with SNPs F_{ST} >0.1 (n= ~1,900)



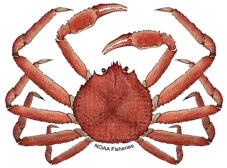
Way less variability in Reduced pH

Fewer heterozygotes (individuals with multiple alleles)

OA Treatment

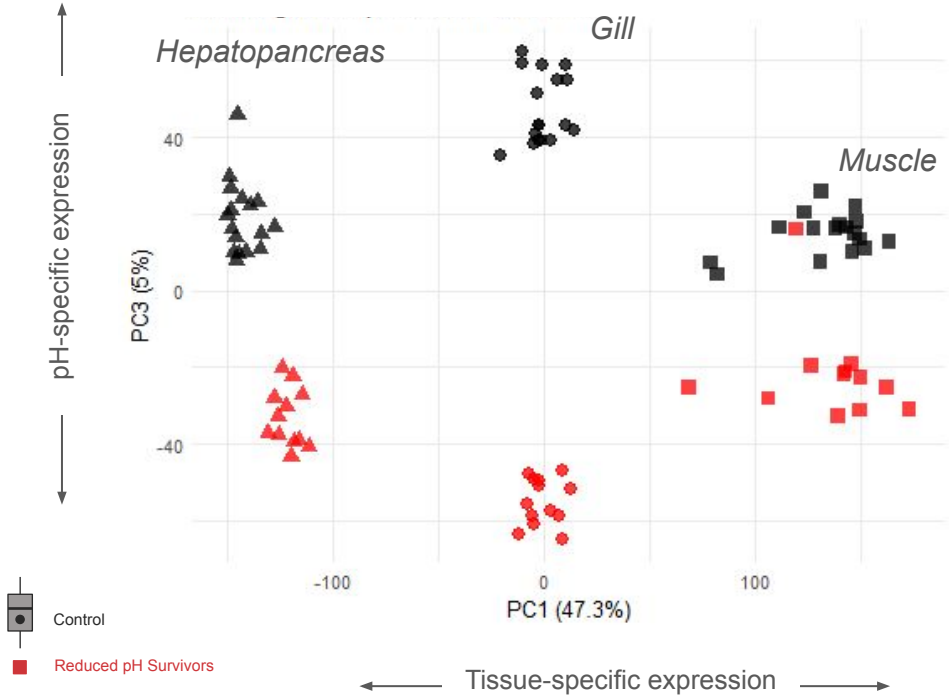
Ambient

Reduced



Some genetic differences (SNPs) are located in expressed genes

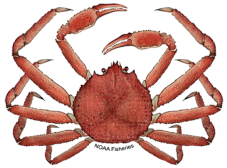
PCA conducted with expressed genes (~11k)



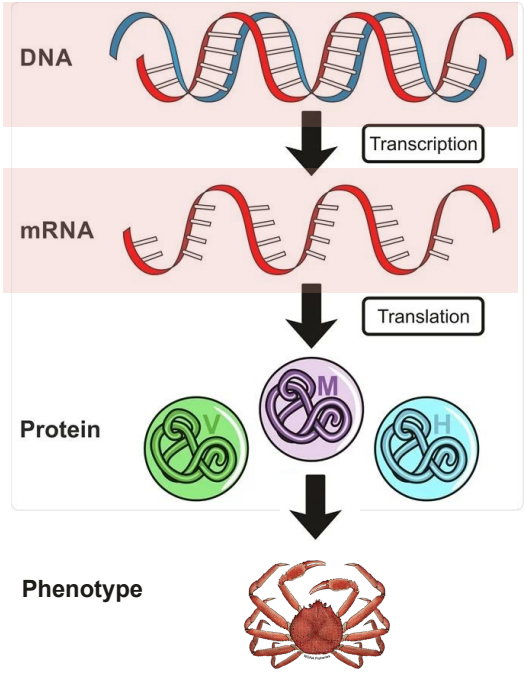
Of 889 SNPs:

- ~25% in known genes
- 11% in genes affected by OA, most of which were in gill & hepatopancreas tissues

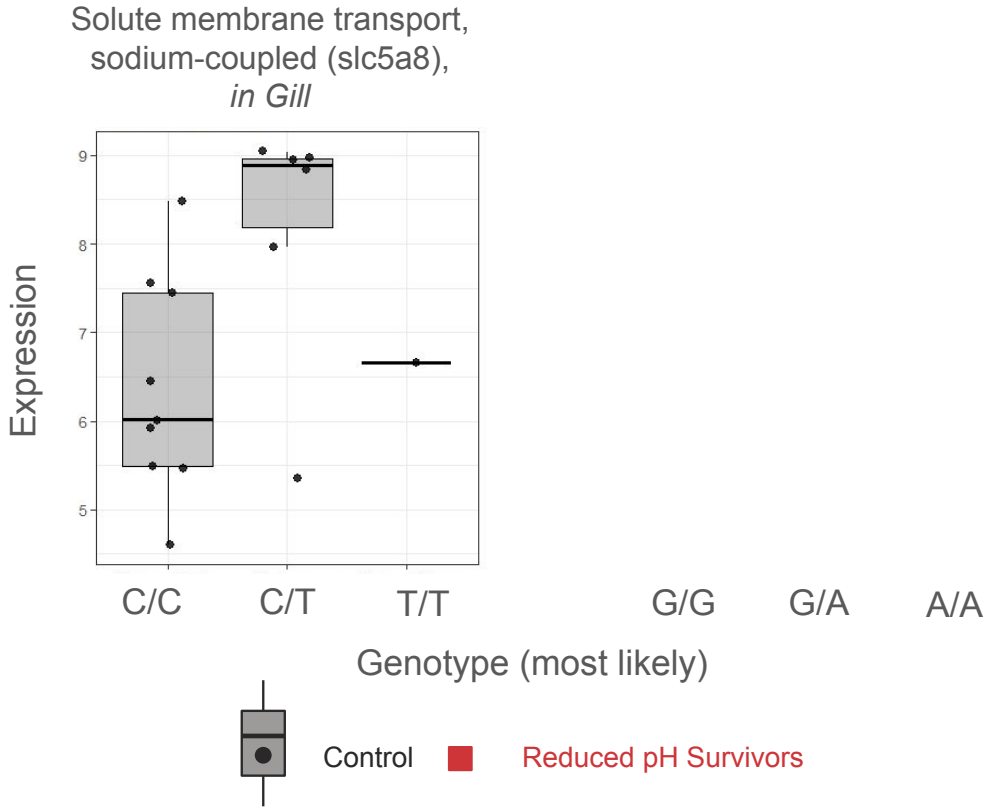
Many (but not all!) of the genes containing these SNPs were involved in transmembrane transport

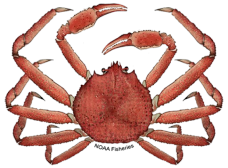


Some genetic differences may influence gene expression

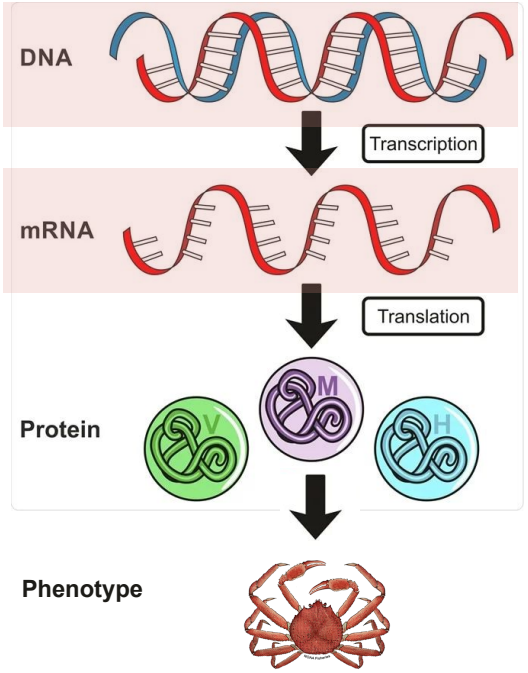


Adapted from: udaix/Shutterstock.com



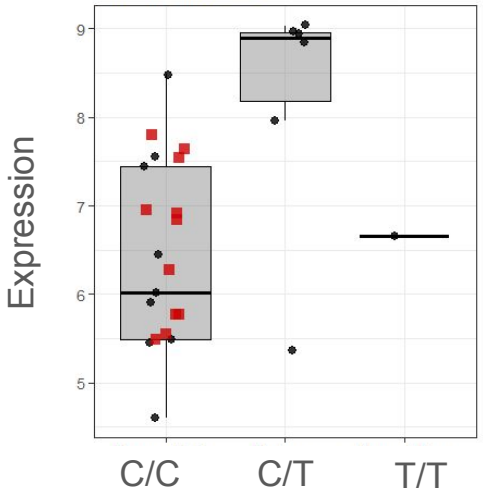


Some genetic differences may influence gene expression

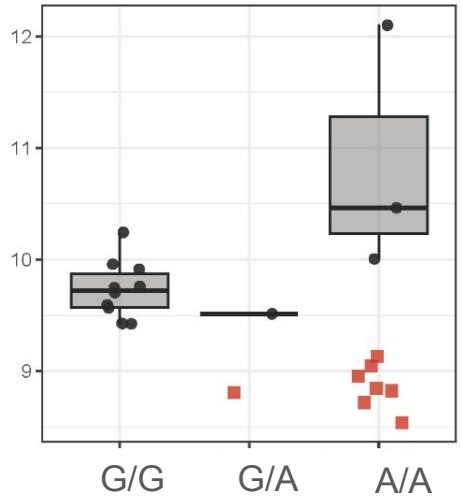


Adapted from: udaix/Shutterstock.com

Solute membrane transport, sodium-coupled (slc5a8), *in Gill*



Electron transport and ATP synthesis (NDUFB8), *in Hepatopancreas*

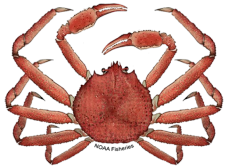


Genotype (most likely)



Control

Reduced pH Survivors



What is distinct about Tanners that survive in OA?

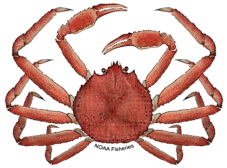
1. OA survivors have less genetic variation where there are genetic differences (but, very small subset of SNPs).

2. ~25% of genetic differences are in annotated genes* which are enriched for transmembrane transport processes, primarily solute transport.

   **Genome is poorly annotated, but I'm working on that!*

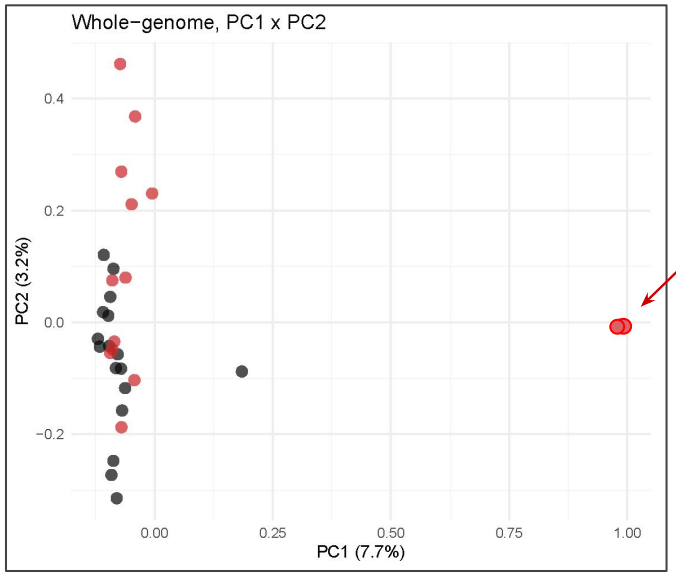
3. Most SNPs in OA-responsive genes (DEGs) were in gill tissue. Gill function likely critical for OA tolerance.

One more thing …



Outliers in PCA, genetic data (~350k SNPs)

Tanner crab selection study

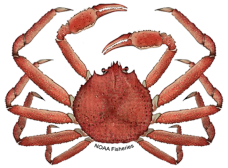


????

OA Treatment

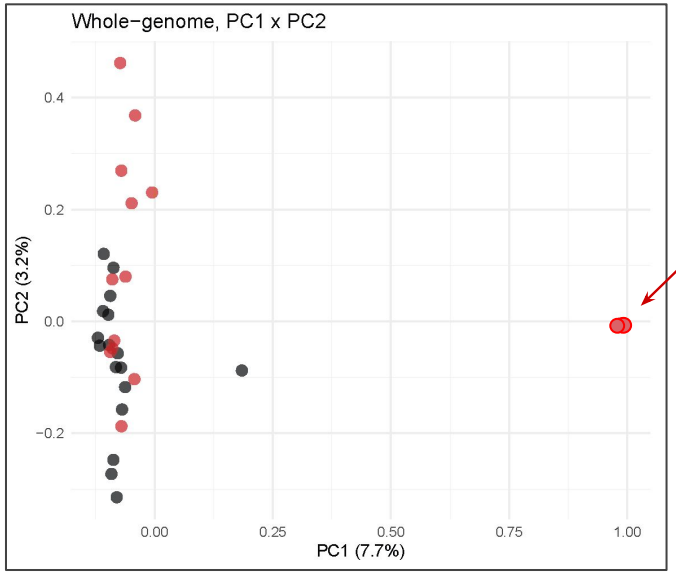
Ambient

Reduced



Outliers in PCA, genetic data (~350k SNPs)

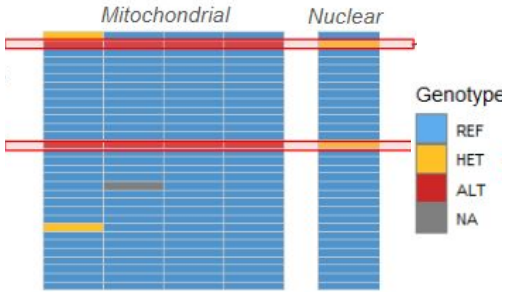
Tanner crab selection study



OA Treatment

- Ambient
- Reduced

Hybrids! *Two of 14 OA survivors*

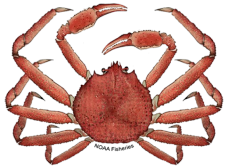


SNP that differ among tanner crab (REF) and snow crab (ALT)

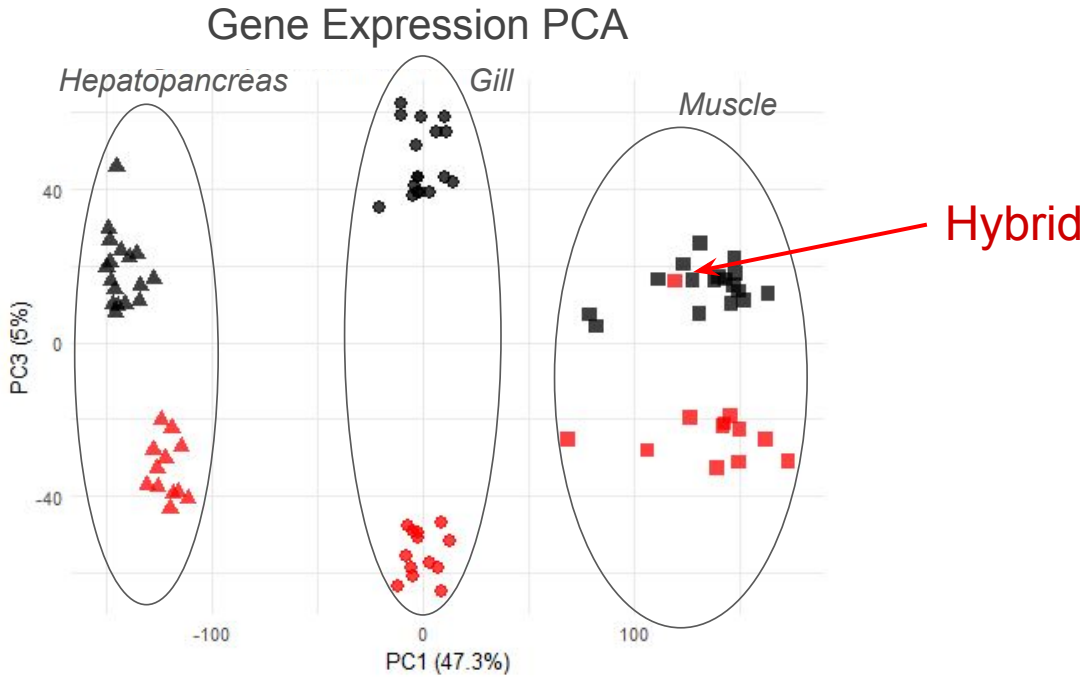
lcWGS mapped to nuclear & mito sequences:

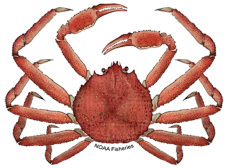
- *16S: mitochondrial, maternal &*
- *ITS: nuclear, both parents*

(Smith, Grant, and Seeb, 2005)



A hybrid's muscle gene expression profile looks like control crab

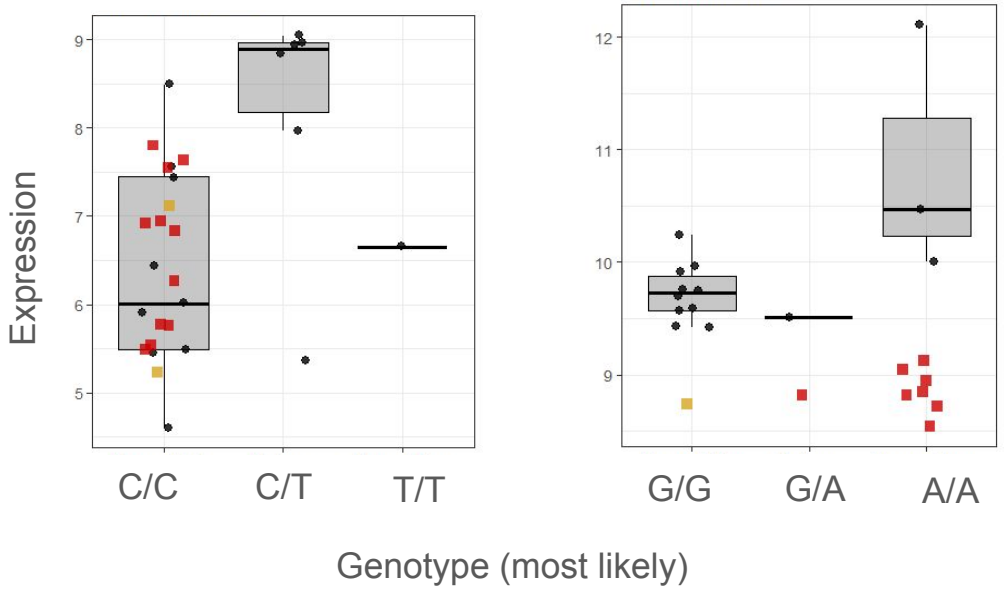




In some cases, hybrid genotype and expression profile matches survivors

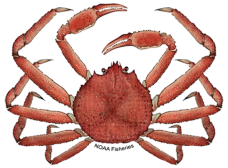
Solute membrane transport, sodium-coupled (slc5a8), *in Gill*

Electron transport and ATP synthesis (NDUFB8), *in Hepatopancreas*



Control

Reduced pH Survivors



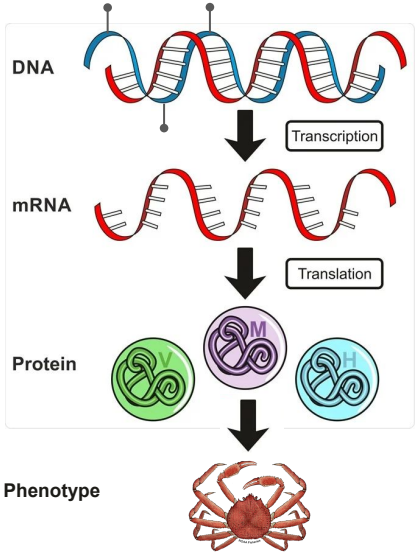
Are OA survivors more genetically similar to hybrids?

I'm still analyzing, but …

- No consistent evidence that Tanner OA survivors are more genetically similar to hybrids based on
    a. Prevalence of snow crab alleles
    b. Mapping depth to snow crab genome
    c. % of genome covered
    d. Pairwise genetic distances
    e. Kinship (KING) higher among OA survivors & hybrids, compared to Control and hybrids (?)

Are hybrids more OA tolerant?
Can hybrids transfer OA tolerance to Tanners by interbreeding?

Growing crab ‘Omics resources



Adapted from: udaix/Shutterstock.com

- Improved Snow crab genome annotations
- Tanner genome assembly
- Tissue-specific gene expression data, some paired with genetics
- Sequences for candidate biomarkers of OA & other stressors (e.g. HSPs)
- Hoping to fund epigenetic aging pilot project

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